



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL

General Certificate of Education Advanced Level

FURTHER MATHEMATICS

9187/1

PAPER 1

NOVEMBER 2013 SESSION

3 hours

Additional materials:

Answer paper

Graph paper

List of Formulae

TIME 3 hours

INSTRUCTIONS TO CANDIDATES

Write your name, Centre number and candidate number in the spaces provided on the answer paper/answer booklet.

There is no restriction on the number of questions which you may attempt.

If a numerical answer cannot be given exactly, and the accuracy required is not specified in the question, then in the case of an angle it should be given to the nearest degree, and in other cases it should be given correct to 2 significant figures.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question.

The total number of marks for this paper is 120.

Questions are printed in the order of their mark allocations and candidates are advised to attempt questions sequentially.

The use of an electronic calculator is expected, where appropriate.

You are reminded of the need for clear presentation in your answers.

This question paper consists of 5 printed pages and 3 blank pages.

Copyright: Zimbabwe School Examinations Council, N2013.

- 1 The points A, B and C have position vectors $\mathbf{a} = 2\mathbf{i} - 3\mathbf{j} + 7\mathbf{k}$, $\mathbf{b} = 3\mathbf{i} - \mathbf{j} + \mathbf{k}$ and $\mathbf{c} = 2\mathbf{i} + \mathbf{j} - 3\mathbf{k}$, respectively.

(i) Calculate $(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c}$ [4]

(ii) Describe fully the geometrical meaning of your calculated value. [2]

- 2 The matrix A is given as

$$A = \begin{pmatrix} 2 & 4 & 0 \\ -1 & 1 & 1 \\ 3 & 5 & 0 \end{pmatrix}.$$

(a) Calculate the determinant of matrix A, [1]

Hence or otherwise, find the inverse of matrix A. [5]

(b) When point B (a, b, c) is transformed by matrix A its image is C ($4, -1, 6$).

Find the values of a, b and c . [2]

- 3 (i) Write $9x^2 - 24x + 10$ in the form $(ax + b)^2 + c$, where a, b and c are all integers to be found and $a > 0$. [2]

(ii) Hence from (i) or otherwise, show that

$$\int_3^4 \frac{1}{\sqrt{9x^2 - 24x + 10}} dx = \frac{1}{3} \ln \left(\frac{8 + \sqrt{58}}{5 + \sqrt{19}} \right). \quad [7]$$

- 4 (i) Given that $I_n = \int_0^{\frac{\pi}{4}} \sin^n 2x dx$, show that

$$I_n = \left(\frac{n-1}{n} \right) I_{n-2}. \quad [6]$$

(ii) Hence from (i) or otherwise, evaluate

$$\int_0^{\frac{\pi}{4}} \sin^4 2x dx. \quad [3]$$

5

A set $G = \{a, b, c, d\}$ has its elements defined using the turning of a minute hand of a clock as follows:

a	=	turn of 0 or multiples of 60 minutes,
b	=	turn of 15 minutes,
c	=	turn of 30 minutes
d	=	turn of 45 minutes

The multiplication rule $b * c = d$ means "when turn b is followed by turn c " the result is equivalent to a single turn of d .

- (i) Construct a multiplication table for the elements.

Show that the table forms a group.

[7]

- (ii) Find the order of each element.

[3]

- (iii) Write down a subset of G , which forms a group on its own.

[1]

- 6 (a) The curve C has equation $y = ax^{\frac{3}{2}}$, where a is a constant.

- (i) Show that the length, L , of the arc of the curve, C , between $x = 0$ and $x = 1$ is given by

$$L = \frac{8}{27a^2} \left\{ \left(1 + \frac{9a^2}{4} \right)^{\frac{3}{2}} - 1 \right\}.$$

- (ii) Hence show that when a is a large number, $L \approx a$.

[7]

- (b) The arc of the curve $y^2 = x$ between $x = \frac{3}{4}$ and $x = 6$, is rotated about the x -axis.

Find the value of the surface area generated.

[5]

7 The function $f(x)$ is given by $f(x) = \frac{5x^2 + 4x + 7}{(3x+1)(x^2+3)}$.

(i) Express $f(x)$ in partial fractions. [4]

(ii) Hence show that

$$\int_0^1 f(x) dx = \frac{7}{6} \ln 4 - \frac{1}{2} \ln 3 + \frac{1}{\sqrt{3}} \tan^{-1} \left(\frac{1}{\sqrt{3}} \right). \quad [4]$$

(iii) Find the maclaurin series expansion of $f(x)$ up to and including the term in x^2 . [4]

8 (a) (i) Given that when $x = \pi$, $y = -2$, solve the differential equation

$$\frac{dy}{dx} + y \tan x = \sin 2x, \text{ expressing } y \text{ in terms of } x.$$

(ii) Sketch the curve of y in (i) against x for $0 \leq x \leq 2\pi$. [8]

(b) Find the general solution of the differential equation

$$6 \frac{d^2 y}{dx^2} + 5 \frac{dy}{dx} - 6 = 0. \quad [4]$$

9 (a) Factorise $4x^2 - 1$. [1]

(b) The function $f(r)$ is given by

$$f(r) = \frac{16r^2 + 32r + 28}{(4r^2 - 1)(2r + 3)(2r + 5)}.$$

(i) Reduce $f(r)$ to partial fractions.

(ii) Hence, or otherwise, find as a single fraction in terms of n , the sum of the series

$$\sum_{r=1}^n f(r). \quad [11]$$

10 The function $f(x)$ has equation given as $f(x) = \frac{1}{12}(x^4 - 6x^2 + 24)$.

(i) Find the coordinates

1. of the turning points and determine their nature,
2. of the points of inflexion.

[9]

(ii) Sketch the curve of the function $y = -\frac{1}{f(x)}$.

NB Your sketch must show coordinates of all the turning points and the points of inflexion, and asymptote line(s), if any.

[4]

11 The plane p has equation $r \cdot (i - j + k) = 3$, and the plane π passes through the points A (2, -3, 10), B(-2, 7, 4) and C(4, 3, 2).

(i) 1. Find the vector equation of the plane π .

2. Hence find the vector equation of l , the line of intersection of the planes p and π .

[7]

(ii) A point M has coordinates (-6, -6, 3).

Show that the point M lies on plane p but does not lie on plane π .

[3]

(iii) The point N lies on the line l such that NM is perpendicular to the line l .

Find, the coordinates of N.

[3]

(iv) Find the distance of the point M from the plane π .

[3]